Assessment Task No 1			
Topic:	**Introduction to Machine Learning**	**Week No.**	**2-3**
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Student Name	Mrincess Allisha C. Sombrano	**Section**	BSCS-3B
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Performance Task 1: From Data to Prediction — Analyzing Real-World Machine Learning Applications

Part A — Identify the ML Problem

Question	Your Answer
1. What is the application?	Content/Video Recommendation System
2. What problem are you trying to solve?	The problem is how to recommend videos or content that a user is likely to enjoy or watch based on their previous activities and interests.
3. What are the inputs?	The inputs can include the user's watch history, likes, dislikes, searches, video categories, watch time, and interactions with videos.
4. What is the expected output?	The expected output is a list of recommended videos or content that the user is likely to watch or enjoy.
5. What patterns or relationships might the ML system learn?	The ML system might learn that users who watch, like, or search for certain types of videos are more likely to enjoy similar content. For example, if a user frequently watches programming tutorials, the system may learn to recommend more programming-related videos.
6. What data would the system need?	The system would need data such as users' watch history, liked and disliked videos, search history, watch duration, video categories, ratings, and interactions such as clicks and shares.

Part B — Traditional Programming vs Machine Learning

Traditional Programming
Input -> Programmer's Algorithm/Rules -> Output

➔ In Traditional programming, the programmer tells the computer exactly what to do by creating rules.

User watches coding video → Programmer created the rule: *"If the user watches coding videos, recommend another coding video."* → Recommend another coding video

The computer is following a rule that was already written by the programmer.

Machine Learning
Input/Data → ML Algorithm → Learned Pattern → Output/Prediction

➔ In Machine Learning, instead of the programmer creating every possible rule, the computer is given data and learns patterns from it.

Watch history + likes + searches + watch time → ML studies the user's behavior → Learns**:** "This user seems to like coding and programming videos." → Predicts and recommends videos the user may like

The computer learns from examples instead of simply following manually written rules.

Part C — Create Your Own Mini Dataset

Input	**Output**
1. Watches and likes sports videos	Recommends more sports videos
2. Watches cooking videos	Recommends more cooking videos
3. Likes gaming videos	Recommends more gaming videos
4. Search coding tutorial	Recommends more programming videos
5. Spend time watching vlog	Recommends more vlog videos

1. Identify the pattern.

➔ The pattern is that the system recommends videos based on the user's interests and viewing behavior. If the user watches, likes, or searches for a certain type of content, the system learns that the user may be interested in similar content.

2. Determine the missing output.

➔ For example:

Input: Watches and likes sports videos
Output: Recommends more sports videos

So, the missing output should be more videos related to the user's interests.

3. Write the mathematical relationship/function.

➔ Recommended Videos = f(User's Interests and Viewing Behavior)

➔ User Behavior → Similar Content Recommendation

4. Explain how a machine learning algorithm could learn this relationship from the examples

➔ The ML algorithm looks at many examples of what users watch, like, search for, and spend time watching. From these examples, it identifies patterns between user behavior and the type of videos they are interested in. Then, when a user shows similar behavior, the algorithm predicts what type of videos they might like and recommends them.

Part D — Explain the Training Process

In 3—5 sentences, answer:

How does a Machine Learning algorithm learn from data?

➔ A Machine Learning algorithm receives input data, such as a user's watch history, likes, and searches, along with the expected output, such as the videos the user may like. During training, the algorithm studies many examples and looks for patterns between the input and output. It learns the relationship between the user's behavior and the type of content they prefer. After learning, the algorithm can use new input data to make a prediction, such as recommending a video the user may enjoy.

Reflection:
Why is learning from data useful instead of manually programming every possible rule?

➔ Learning from data is useful because there are many different users with different interests and preferences, so it would be difficult to manually program every possible rule. For example, one user may enjoy funny vlogs, while another may prefer gaming, cooking, or sports videos. Instead of creating a separate rule for every user, the ML system can study large amounts of data and learn patterns from users' behavior. It can use these patterns to make better and more personalized recommendations. As more data becomes available, the system can continue learning and improving its predictions.